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| **Duration** | 4 mins 3 secs |
| **Grade** | **13.00** out of 15.00 (**86.67**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It is based on the divide-and-conquer approach
c. It sorts data using a single pass
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **2**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Dividing the list into halves
c. Bubbling up the largest element to its correct position
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **3**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Bubble sort

c. Insertion sort

d. Heap sort

The correct answer is:
Insertion sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()

b. order()

c. sort()

d. arrange()

The correct answer is: sort()

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process

b. It makes it easier to select items based on their relationship to the rest of the items

c. It makes the data unsorted

d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 0.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- b. A sorting algorithm is stable if it preserves the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **8**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are ignored
- b. Each part is searched for a specific element
- c. Each part is sorted independently
- d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting decreases the efficiency of selection operations
c. Sorting helps in finding duplicates quickly
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data sequentially
c. Sorting data in a single pass
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Arranging data in a specific order
c. Merging two datasets
d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It only works with integer arrays
c. It sorts data out of the box efficiently
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Dynamic programming
b. Greedy method
c. Divide and conquer
d. Backtracking

The correct answer is:
Divide and conquer

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data
b. It avoids the need for recursion
c. It only works on small datasets
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **15**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. sort()
- c. arrange()
- d. order()

The correct answer is: sorted()